

Paper 1: Automatic Flood Detection from Sentinel-1 Data Using Deep Learning Architectures

Authors: B. Ghosh, S. Garg, M. Motagh (2022)

The study addresses the real time need for automated, near real time flood pattern mapping to support disaster management authorities. The authors contrast their approach from traditional binary segmentation methods, such as threshold splitting. The backscatter across different geographical regions is an essential drawback for flood mappings through traditional methods. The paper proposes use of two deep learning architectures, UNet and Feature pyramid network (FPN).

The models were trained and validated using the NASA Sen1Floods11 dataset, which has varying flood event data from regions such as Nebraska, Bangladesh, and Alabama. The Unet and FPN show a mean intersection over union of 75.06% and the FPN reaching 75.76%. The paper concludes by assessing that deep learning models can generalize results of specific data sets, even when trained on data sets belonging to other regions.

Paper 2: A Fusion Approach for Flood Mapping Using Sentinel-1 and Sentinel-2 Datasets

Authors: Beste Tavus, Sultan Kocaman, Hakan A. Nefeslioglu, Candan Gokceoglu (2020)

The paper proposes a feature level fusion approach that combines Sentinel-1 SAR and Sentinel-2 optical data to map flood extents more effectively. The authors note that optical data is easier to interpret, in spite of it being hindered by cloud coverage during flood events. This mandates expert processing. The paper utilizes a random forest (RL) supervised classifier to integrate features like polarimetric decomposition (H-Alpha) from SAR and water indices (NDVI, MNDWI) from optical imagery.

Consensus results from 2018 flood monitoring in ordu, turkey show that Sentinel-2 data significantly aids in training sample selection and settlement extraction, while Sentinel-1 compensates for cloud-covered areas. The study concludes that multi sensor fusion strategy is superior for rapid flood pattern assessment, offering an efficient approach that leverages the strengths of both sensor types.

Paper 3: Data Processing Architectures for Monitoring Floods Using Sentinel-1

Authors: W. Wagner et al. (2020)

The study investigates the optimal system design for automated global flood monitoring using the extensive time series data provided by the sentinel 1. The authors categorize the data processing systems into three types, single image, dual image and data cube architecture. The study highlights that single image is prone to errors from wind induced waves or low contrast mineral/rock bodies like asphalt. Advanced architectures can be used to overcome these confounding effects.

The authors strongly recommend data cube architectures for large scale data mappings, through higher compute and storage. They enable advanced algorithms like change detection and seasonal trend modelling, which are relatively less sensitive to false positives. Austrian data cube was used to demonstrate how this approach provides high quality flood maps that include extensive data such as retrieval uncertainty and exclusion masks for urban or forested regions, which offer much less radar detection.

Paper 4: DeepFuse: Towards Frequent Flood Inundation Monitoring using AI and EO

Authors: A. Dasgupta, et al. (2024)

In this paper I identified this that the team behind DeepFuse noticed a major problem: the best satellites for detailed flood mapping only fly over a specific spot every few days. This means they

could easily miss the most critical moment when the flood water is at its highest. The picture they

get might show the flood starting or ending, but not the peak, which is the most important information for disaster response.

To solve this, they came up with a clever workaround. They trained a neural network to act like a smart simulator. The AI learned the relationship between two things: the frequent but coarser data

we have every day (like soil moisture and rainfall estimates) and the rare but detailed satellite images. By feeding the model the daily conditions, it could then predict what a high-resolution flood

map would look like on that day, even without a satellite pass.

They tested this idea in France and it worked really well. The AI was able to fill in the gaps between

satellite images, creating a near-daily picture of how the flood evolved. This "temporal fusion"

approach means we don't have to wait for a satellite to be overhead to understand what's happening on the ground, giving us a much more complete and timely view of a developing flood.

Paper 5: A GAN-Based SAR-Optical Data Fusion Approach for Enhanced Flood Mapping

Authors: T. Sekine, et al. (2024)

Optical images are clear and easy to understand, but clouds always seem to cover the sky right when

a flood happens. Radar can see through clouds just fine, but the images it produces are grainy and

hard to interpret, which can lead to mistakes in flood maps. The researchers wondered, "What if we

could teach the radar to see more clearly using the optical data?"

They came up with a really clever solution using a type of artificial intelligence called a Generative

Adversarial Network, or GAN. Think of it like this: they used the clear optical images as a perfect teacher to guide the messy radar data. The GAN learned how to take a noisy radar image and transform it into a much cleaner version that still kept its all-weather capability. It was like giving the

radar data a pair of glasses made from optical images.

By fusing the two data types in this smart way, the model produced much more accurate and detailed flood maps. It successfully combined the best of both worlds: the reliability of radar in any

weather with the rich detail and clarity of optical imagery. This approach shows that one sensor can

actually help another "see" better, leading to more trustworthy results during emergencies.

Paper 6: Vision Transformer for Flood Detection Using Satellite Images from Sentinel-1 and Sentinel-2

Authors: K. Papoutsis, et al. (2024)

In this paper I identified this that the researchers decided to try something new for flood detection.

Instead of using the standard deep learning models that have been popular for years, they used a

Vision Transformer (ViT). This type of model was originally designed to understand language and text,

but the researchers cleverly adapted it to "read" satellite images like sentences, understanding how

different parts of the image relate to each other.

They tested this smart model separately on two types of satellite data: radar images from Sentinel-1

(which can see through clouds) and optical images from Sentinel-2 (which gives us clear, natural-looking pictures but struggles with cloud cover). The goal was simple: could the model just look at an

image and quickly say "yes, this is a flood" or "no, this is not a flood"? This would act like a super-fast

first alert system.

The results were pretty impressive. The Vision Transformer achieved around 84% accuracy on the

tricky radar data and 83% on the optical data. More importantly, it was much better than older models, beating them by up to 15% on the radar images. This tells us that these transformer models

are really good at understanding different kinds of satellite pictures and could be a powerful tool for

building fast and reliable flood warning systems in the future.

Paper 7: Multimodal and Multitemporal Data Fusion for Flood Extent Segmentation

Authors: L. Gomez-Chova, et al. (2024)

This research focuses on creating a more robust and accurate flood mapping system by combining everything: multiple types of data (radar and optical) from multiple points in time (before and after the flood). The team used two major flood datasets (Kuro Siwo and WorldFloods) to train deep learning models. By feeding the models images from both Sentinel-1

and Sentinel-2 taken before and during a flood, the AI learned to identify the changes specifically caused by the flood event itself, ignoring permanent water bodies and other non-flood-related features. Their best fusion model achieved a 70% overlap with hand-drawn maps, which was a significant 10% improvement over models that only used one type of data. This shows that a combined approach, looking at all available information, is key for reliable flood mapping.

Paper 8: A framework for flood inundation extraction based on microwave and optical remote sensing images.

Authors: Teng Q, Luo L, Li S, Xing L, Shao K, Wang S and Wang D (2024)

Remote sensing has emerged as a critical technology for large-scale flood monitoring because of its capability to provide synoptic, repetitive, and objective observations. Traditional hydrodynamic simulation models, although useful, often require high-quality ground data and substantial computational resources, limiting their rapid deployment during emergencies.

Recent studies emphasize the importance of satellite-based monitoring systems for near real-time flood mapping and disaster response. The increasing availability of high-resolution satellite platforms, particularly from the Sentinel constellation, has enabled continuous observation of flood-prone regions. However, the rapid growth in microwave and optical telemetric data has introduced challenges related to preprocessing complexity, noise removal, calibration, and computational efficiency—especially when datasets are unsupervised and large in scale.

The study by Sentinel-2 demonstrates the effectiveness of multispectral imagery for pre-flood water body delineation. Optical data allow precise discrimination between water and land surfaces under clear-sky conditions. Methods such as single-band thresholding, supervised classification, support vector machines, and hybrid water index models have significantly improved extraction accuracy.

The reference framework employs Sentinel-1, which provides C-band SAR data with high temporal resolution. Water surfaces typically produce low backscatter values due to specular reflection, enabling threshold-based classification methods for flood extraction.

Common SAR-based techniques include:

Otsu thresholding

Object-oriented segmentation

Fuzzy C-means (H-FCM)

Deep learning models (e.g., U-Net, contrastive learning approaches)

Paper 9: Monitoring of flood water propagation based on microwave and optical imagery

Authors: Nusrat Jahan Ety a, Zhongyi Chu b, Syed M. Masum a

The paper studies how microwave and optical satellite imagery can be combined to monitor and analyse flood water propagation in flood-prone regions. The paper focuses on the region Dinajpur in Northern Bangladesh. The paper addresses issues such as the cost, pace and the limitation of the traditional methods like ground surveys and aerial observation. Optical imagery provides detailed visual information but is constrained by clouds and haze during flood events. Microwave Synthetic Aperture Radar (SAR) can penetrate clouds and operate day/night that can be used for flood monitoring under adverse conditions. The data was acquired from missions supported by the European Space Agency. Result shows that multi sensor integration increases the mapping reliability and that the model is scalable to other regions as it relies on the freely available satellite data.

Paper 10: FLOOD MONITORING USING SENTINEL-1 SAR DATA: A CASE STUDY BASED ON AN EVENT OF 2018 AND 2019 SOUTHERN PART OF KERALA

Authors: S. P. Dhanabalan, S. Abdul Rahaman, and R. Jegankumar

The paper focuses on the southern Kerala region in India which is a flood prone region due to its terrain characteristics. The paper investigates the effectiveness of radar based remote sensing for mapping flood in the region. The author discusses the limitation of the conventional field surveys and method such as the cost, time required. The author employs C-band Synthetic Aperture Radar imagery from European Space Agency's Sentinel-1A mission which valuable in flood scenarios as it can operate independently in daylight and can penetrate cloud cover which is advantageous during monsoon periods when optical sensors are frequently obstructed. The author's findings demonstrates that SAR analysis supports rapid disaster assessment and recovery planning. The authors emphasize that the model can be scaled because Sentinel data are freely accessible and thus can be used in resource-constrained regions.

References:

1. Automatic Flood Detection Using Deep Learning on Sentinel-1 Data
Author: B. Ghosh, S. Garg & M. Motagh

2. *A Fusion Approach for Flood Mapping Using Sentinel-1 & Sentinel-2*
Author: B. Tavus et al
3. *Data Processing Architectures for Monitoring Floods Using Sentinel-1*
Authors: Wagner et al.
4. *DeepFuse: Towards Frequent Flood Inundation Monitoring using AI and EO*
Authors: A. Dasgupta, et al. (2024)
5. *A GAN-Based SAR-Optical Data Fusion Approach for Enhanced Flood Mapping*
Authors: T. Sekine, et al. (2024)
6. *Vision Transformer for Flood Detection Using Satellite Images from Sentinel-1 and Sentinel-2*
Authors: K. Papoutsis, et al. (2024)
7. *Multimodal and Multitemporal Data Fusion for Flood Extent Segmentation*
Authors: L. Gomez-Chova, et al. (2024)
8. *A framework for flood inundation extraction based on microwave and optical remote sensing images.*
Authors: Teng Q, Luo L, Li S, Xing L, Shao K, Wang S and Wang D (2024)
9. *Monitoring of flood water propagation based on microwave and optical imagery*
Authors: Nusrat Jahan Ety a, Zhongyi Chu b, Syed M. Masum a
10. *FLOOD MONITORING USING SENTINEL-1 SAR DATA: A CASE STUDY BASED ON AN EVENT OF 2018 AND 2019 SOUTHERN PART OF KERALA*
Authors: S. P. Dhanabalan, S. Abdul Rahaman, and R. Jegankumar